Speech Acts and Embedded Sentences in Logical Form
A Detailed Explanatory Chapter

Introduction

Natural language sentences do not merely describe the world; they perform actions. When a speaker utters a sentence, they are not only expressing propositional content but also performing a speech act: asserting, questioning, commanding, promising, warning, or requesting. The distinction between propositional content and illocutionary force is foundational in the philosophy of language and formal semantics.

At the same time, sentences may occur embedded inside larger constructions, such as belief reports, questions, or modal contexts. Embedded clauses often lose their independent speech act force and instead function as propositional arguments. Understanding how speech acts interact with embedding is essential for constructing a precise logical form.

This chapter explains how speech acts are represented formally, how illocutionary force differs from propositional content, and how embedded clauses are encoded in logical form.

1. Speech Acts: Propositional Content vs Force

Following Austin and Searle, we distinguish:

- **Locutionary act**: the production of a linguistic expression.
- **Illocutionary act**: the communicative force (asserting, asking, commanding).
- **Perlocutionary act**: the effect on the hearer.

In formal semantics, we focus on illocutionary force. A declarative sentence like:

$$Rina\ left.$$

has propositional content:

$$\exists e(Leave(e) \land Agent(e, r))$$

But when uttered, it performs an assertion. We may represent this using an assertion operator:

$$ASSERT(\exists e(Leave(e) \land Agent(e, r)))$$

The operator ASSERT applies to a proposition. This separates force from content.

2. Declaratives, Interrogatives, and Imperatives

2.1 Declaratives

Declaratives express propositions.

$$Rina\ left.$$

Logical form:

$$ASSERT(\exists e(Leave(e) \land Agent(e, r)))$$

The proposition is truth-evaluable; ASSERT marks the speech act.

2.2 Interrogatives

Questions do not assert truth but request information.

$$Did\ Rina\ leave?$$

The propositional content is:

$$\exists e(Leave(e) \land Agent(e, r))$$

The force operator is:

$$QUESTION(\exists e(Leave(e) \wedge Agent(e, r)))$$

For wh-questions:

$$Who\ left?$$

The logical representation involves abstraction:

$$\lambda x \exists e(Leave(e) \wedge Agent(e, x))$$

The QUESTION operator applies to a set of propositions.

2.3 Imperatives

Imperatives express directives.

$$Leave!$$

Logical content:

$$\lambda x \exists e(Leave(e) \wedge Agent(e, x))$$

Force operator:

$$COMMAND(\exists e(Leave(e) \wedge Agent(e, hearer)))$$

Imperatives do not describe but attempt to change the world.

3. Embedded Declarative Clauses

Consider:

$$Rina\ believes\ that\ Amit\ left.$$

The embedded clause "Amit left" has propositional content:

$$\exists e(Leave(e) \land Agent(e, a))$$

But it is not asserted by the speaker. Instead, it is the object of belief.

Logical form:

$$Believes(r, \exists e(Leave(e) \land Agent(e, a)))$$

Notice that ASSERT does not apply to the embedded clause. The clause contributes content but not independent speech act force.

4. Intensionality and Embedded Clauses

Attitude verbs like "believe," "want," and "know" create intensional contexts.

Example:

Rina believes that the winner cheated.

De dicto reading:

$$Believes(r, \exists x(Winner(x) \land Cheat(x)))$$

De re reading:

$$\exists x(Winner(x) \land Believes(r, Cheat(x)))$$

The ambiguity arises from scope relations between quantifiers and attitude operators.

5. Embedded Questions

Consider:

$$Rina\ asked\ who\ left.$$

The embedded wh-clause denotes a set of propositions:

$$\lambda x \exists e(Leave(e) \wedge Agent(e, x))$$

Logical form:

$$Ask(r, \lambda x \exists e(Leave(e) \wedge Agent(e, x)))$$

The embedded clause contributes interrogative content but lacks independent QUESTION force.

6. Force Deletion Under Embedding

A key principle is that embedded clauses typically lack independent illocutionary force.

Independent clause:

$$ASSERT(p)$$

Embedded clause:

$$Believes(r, p)$$

The absence of ASSERT inside the complement explains why embedded clauses are not independently asserted.

7. Compositional Derivation Example

Sentence:

$$Rina\ believes\ that\ Amit\ left.$$

Lexical entries:

$$left = \lambda x \lambda e(Leave(e) \wedge Agent(e, x))$$

$$believes = \lambda p \lambda x Believes(x, p)$$

Step 1: Compose embedded clause:

$$(\lambda x \lambda e...)(a)$$

$$= \lambda e(Leave(e) \wedge Agent(e, a))$$

Step 2: Existential closure:

$$\exists e(Leave(e) \wedge Agent(e, a))$$

Step 3: Apply belief predicate:

$$(\lambda p \lambda x Believes(x, p))(p)$$

$$= \lambda x Believes(x, p)$$

Step 4: Apply subject:

$$Believes(r, p)$$

Where $p = \exists e(Leave(e) \wedge Agent(e, a))$.

8. Restrictions on Embedding

Imperatives generally cannot embed:

$$*\textit{Rina believes leave}!$$

This reflects incompatibility between directive force and propositional embedding.

Similarly, main clause force is typically suppressed in complements.

Conclusion

Speech acts involve both propositional content and illocutionary force. Logical form distinguishes these components by representing force operators separately from propositions. Embedded clauses contribute propositional content but typically lack independent force. Attitude verbs introduce intensional contexts that affect scope and substitution. By separating force from content and encoding embedding compositionally, formal semantics captures the interaction between speech acts and clause structure with precision.